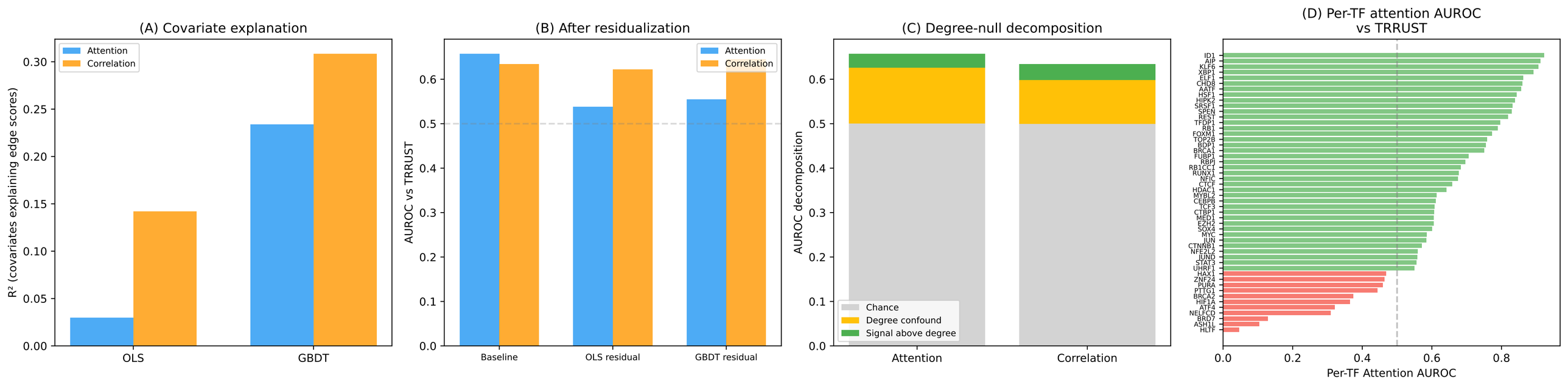
(A) Covariate explanation
Attention
Correlation
R² (covariates explaining edge scores)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
OLS
GBDT
(B) After residualization
Attention
Correlation
AUROC vs TRRUST
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Baseline
OLS residual
GBDT residual
(C) Degree-null decomposition
AUROC decomposition
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Chance
Degree confound
Signal above degree
Attention
Correlation
(D) Per-TF attention AUROC
vs TRRUST
ID1
AIP
KLF6
XBP1
ELF1
CHD8
AATF
HSF1
HIPK2
SRSF1
SPEN
RES1
TFDP1
RB1
FOXM1
TOP2B
BDP1
BRCA1
FUBP1
RBPJ
RB1CC1
RUNX1
NFIC
CTCF
HDAC1
MYBL2
CEBPB
TCF3
CTBP1
MED1
EZH2
SOX4
MYC
JUN
CTNNB1
NFE2L2
JUND
STAT3
UHRF1
HAX1
ZNF24
PURA
PTTG1
BRCA2
HIF1A
ATF4
NELFCD
BRD7
ASH1L
HLTF
0.0
0.2
0.4
0.6
0.8
Per-TF Attention AUROC